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Programa2.Estandarizacion

Estandarización de datos con StandardScaler

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```
In [1]: from sklearn import preprocessing
import numpy as np
```

```
In [2]: # Carga de datos.
X_train = np.array([[ 1., -1.,  2.],
                    [ 2.,  0.,  0.],
                    [ 0.,  1., -1.]])
```

```
In [3]: # Limpieza de datos: estandarización.
standardizer = preprocessing.StandardScaler()
# Estandarizamos los datos de entrenamiento
X_train_std = standardizer.fit_transform(X_train)
print(X_train_std)

[[ 0.          -1.22474487  1.33630621]
 [ 1.22474487   0.          -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

```
In [4]: X_train_2 = np.array([[ 1., -1.,  2.],
                             [ 2.,  0.,  0.],
                             [ 0.,  1., -1.]])
```

```
In [5]: # Aplicamos los métodos "fit" y "transform" por separado
f = preprocessing.StandardScaler()
f = f.fit(X_train)
X_train_2_std = f.transform(X_train_2)
print(X_train_2_std)

[[ 0.          -1.22474487  1.33630621]
 [ 1.22474487   0.          -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

Comprobar si las matrices estandarizadas son iguales

Si `fit_transform` sobre `X_train` y `fit + transform` sobre `X_train_2` (con los mismos datos) dan el mismo resultado, la comparación debe ser `True` .

```
In [6]: # Comprobar si las matrices estandarizadas son iguales  
print(np.array_equiv(X_train_std, X_train_2_std))
```

True